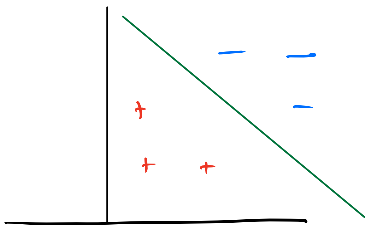
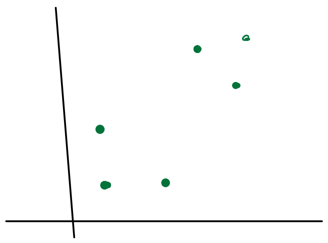
Unsupervised Learning

TODAY: K-means, mixture of Gaussians, EM



Supervised Setting



Unsupervised, no labels!

Unsupervised is harder than supervised

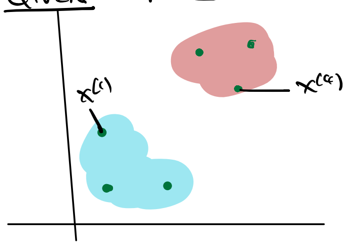
allow Stronger assumptions

accept weaker guarantees

Techniques & ideas are valuable

K-MEANS

Given $K = 2$



Do: 

Given $x^{(1)} \cdots x^{(n)} \in \mathbb{R}^d$ & integer K, # of clusters

Do find assignment of $x^{(i)}$ to one of K clusters

$c^{(i)} = j$ Point i in cluster j

e.g. $c^{(3)} = 2$ while $c^{(4)} = 1$

How do we find these clusters? ITERATIVE Approach

1. Randomly Init $\mu^{(1)}, \mu^{(2)}$

2. Assign each point to closest cluster $\longleftrightarrow$ for each $i = 1 \dots n$ $\quad C^{(i)} = \underset{j=1\dots k}{\operatorname{Argmin}} \|\mu^{(j)} - x^{(i)}\|^2$

3. Compute New cluster centers for $j = 1 \dots k$

$$\mu^{(j)} = \frac{1}{|\Omega_j|} \sum_{i \in \Omega_j} x^{(i)} \quad \text{s.t. } \Omega_j = \{i : C^{(i)} = j\}$$

Repeat until no points change

Comments

Does k-means terminate? Yes!

$$J(c,\mu) = \sum_{i=1}^{n} \|x^{(i)} - \mu_{c^{(i)}}\|^2$$ decreases monotonically

(see notes)

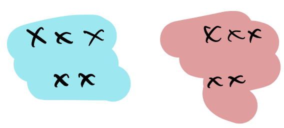
Does it find a global minimum? not necessarily... NP-hard

SIDE NOTE: k-means++ from great Stanford students

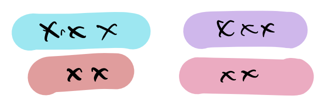
+ Improved Apx Ratio through Clever Init

+ Default in SkLearn

How do you choose k? No ONE Right Answer.



2 clusters



4 clusters

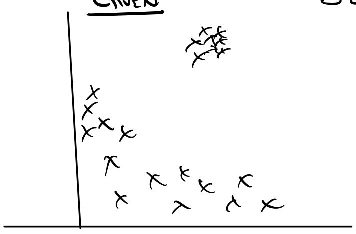
Modeling Question!

Mixture of Gaussians

Toy Astronomy Example (Based on a paper from UW)

- Quasars & stars are sources of light
- Both emit light, and we observe Photons

Given



Do: Assign each photon to light source $P(z^{(i)} = j)$

"Probability Point $z^{(i)}$ belongs to Object j"

cf kmeans. This is a soft assignment

Challenges + Many Sources (say we know k, # of sources)

+ Sources have different intensity & vars

Assume 1. Sources are well modeled by Gaussian (μ_j, σ_j^2)

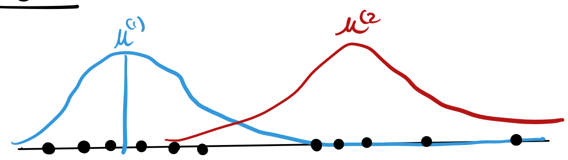
2. We do <u>not</u> assume equal # of points per source

$\rightarrow$ unknown mixture

NB: Physics folks can check if recovered values make sense.

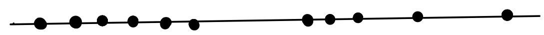
Mixture of Gaussians (model & setup) – 1d for simplicity

Model:

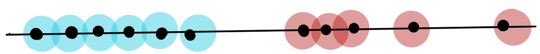


We observe points w/o sources!

$x^{(i)} \in \mathbb{R}$



Observation 1 if we knew "cluster labels" $\longrightarrow$ Solve w/ GDA.



Compute $\mu^{(1)}, \mu^{(2)}$ and be done.

Challenge We don't

Given $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}$ and positive integer k

Do find P s.t. for $i = 1 \dots n$ & $j = 1 \dots k$ clusters

$$P(z^{(i)} = j) \quad \text{soft assignment}$$

According to the "GMM model"

$$P(x^{(i)}, z^{(i)}) = P(x^{(i)} \mid z^{(i)}) P(z^{(i)}) \qquad \text{Bayes Rule}$$

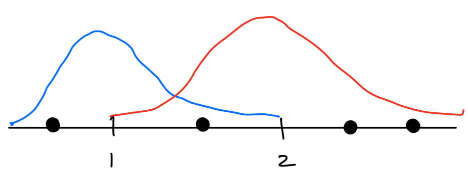
$$z^{(i)} \sim \text{Multinomial}(\Phi) \qquad \Phi_j \geq 0 \quad \sum_{j=1}^{k} \Phi_j = 1 \qquad \text{"which source"}$$

$$x^{(i)} \mid z^{(i)} = j \sim N(\mu_j, \sigma_j^2) \qquad \text{Gaussian in each source}$$

the parameters to be found are highlighted

We call $z^{(i)}$ a hidden or latent variable. $z^{(i)}$ is <u>not</u> directly observed

Example "think sampling"



$\phi_1 = 0.7 \qquad \phi_2 = 0.3$

$\mu_1 = 1 \qquad \mu_2 = 2 \qquad \sigma_1^2 = \sigma_2^2 = \frac{1}{3}$ (roughly)

Repeat
1. Pick blue (w/ prob 0.7) or red (w prob 0.3)
2. Use appropriate mean μ_1 (red) or μ_2 (blue)

GMM Algorithm (Famous Algo & Class)

Mirrors K-means

1. (E-STEP) "Guess latent values" of $z^{(i)}$ for each point
2. (M-STEP) Update parameters

Abstractly our first example of EM-Algorithm (Expectation Maximization)

(E-STEP) <u>Given</u> Data & current guess at parameters $(\phi, \mu, \sigma^2 \dots)$

<u>Do</u> Predict latent variable $z^{(i)}$ for $i = 1..n$

$$w_j^{(i)} = P(z^{(i)} = j \mid x^{(i)}; \phi, \mu, \sigma)$$

<u>Our goal</u>

$$= \frac{P(z^{(i)} = j, x^{(i)}; \phi, \mu, \sigma)}{P(x^{(i)}; \phi, \mu, \sigma)}$$

Bayes Rule

$$= \frac{\overset{①}{P(x^{(i)} \mid z^{(i)} = j; \phi \mu \sigma)} \, \overset{\phi_j}{P(z^{(i)} = j; \phi \mu \sigma)}}{\sum_{l=1}^{k} \underset{②}{P(x^{(i)} \mid z^{(i)} = l; \phi, \mu, \sigma)} \, \underset{\phi_l}{P(z^{(i)} = l; \Phi)}}$$

$* \propto \exp\left\{ - \frac{(x^{(i)} - \mu_j)^2}{\sigma_j^2} \right\}$ "How likely is $x^{(i)}$ according to Gaussian (μ_j, σ_j^2)"

"How likely point from cluster"

<u>Key Point</u> We can compute all terms! Return $w_j^{(i)}$

M-STEP

GIVEN $w_j^{(i)}$ OUR CURRENT ESTIMATE of $P(z^{(i)} = j)$ for $i = 1 \ldots n$
$j = 1 \ldots k$ clusters

DO ESTIMATE OBSERVED PARAMETERS (USING MLE)

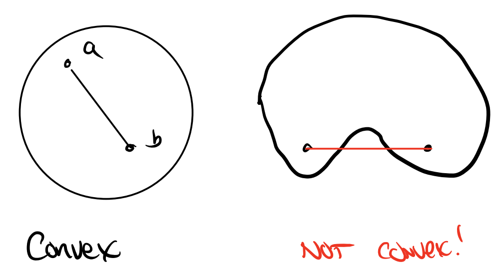
e.g. $\phi_j = \frac{1}{n} \sum_{j=1}^{n} w_j^{(i)}$ $\approx$ fraction of elements in cluster j

$$\mu_j = \frac{\sum_i w_j^{(i)} x^{(i)}}{\sum_i w_j^{(i)}} \quad \ldots \quad \text{etc} \ldots$$

MLE. let's make rigorous!

Detour Convexity & Jensen (This is a key result, we'll go slowly)

A set Ω is convex if for any $a, b \in \Omega$ the line joining a, b is in Ω as well.



In symbols,

$\forall \lambda \in [0,1]$, $a, b \in \Omega$

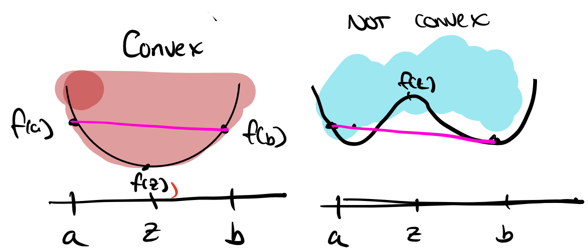
$\lambda a + (1-\lambda) b \in \Omega$

(need to check $\forall a, b \in \Omega$)

Given a function f, the graph of f G_f is defined as

$$G_f = \{ (x,y) : y \geq f(x) \}$$

A function is convex if its graph is convex (as a set)



In symbols, $\forall \lambda \in [0,1]$

$\lambda (a, f(a)) + (1-\lambda)(b, f(b)) \in \Omega$

or let $z = \lambda a + (1-\lambda) b$

$\lambda f(a) + (1-\lambda) f(b) \geq f(z)$

"Every chord is above function"

If f twice differentiable, $\forall x$ $f''(x) \geq 0 \Rightarrow f$ is convex

Pf

$$f(a) = f(z) + f'(z)(a-z) + f''(\xi_a)(a-z)^2 \qquad \xi_a \in [a, z]$$

$$f(b) = f(z) + f'(z)(b-z) + f''(\xi_b)(b-z)^2 \qquad \xi_b \in [z, b]$$

$$\lambda f(a) + (1-\lambda) f(b) = f(z) + f'(z)\big(\cancel{\lambda a + (1-\lambda) b - z}\big)^{0} + c \qquad c \geq 0$$

def of z.

i.e. $\lambda f(a) + (1-\lambda) f(b) \geq f(z)$ $\square$

We say f is strongly convex if $\forall x \in \mathrm{Dom}(f)$ $f''(x) > 0$.

ex: $f(x) = x^2 \Rightarrow f''(x) = 2 \Rightarrow$ strongly convex

$f(x) = x^2(x-1)^2$: graph above (not convex)

Jensen's Inequality $\mathbb{E}[f(x)] \geq f(\mathbb{E}[x])$ for convex f.

ex: X takes value a with prob λ

takes value b with prob $1-\lambda$

$\mathbb{E}[f(x)] = \lambda f(a) + (1-\lambda) f(b)$

$f(\mathbb{E}[x]) = f(z)$ $z = \lambda a + (1-\lambda) b$

NB: can prove finitely supported distribution by induction

for convex f, definition implies thm in this case!

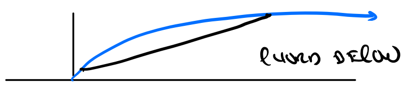
Stronger if f is strongly convex, and $\mathbb{E}[f(x)] = f(\mathbb{E}[x])$

$\Rightarrow X$ is a constant (except: almost surely)

We need concave functions g concave iff $-g$ is convex

$\mathbb{E}[g(x)] \leq g(\mathbb{E}[x])$

ex: $g(x) = \log(x) \Rightarrow g''(x) = -x^{-2}$ on $(0,\infty)$ negative



What about $h(x) = ax + b$ convex + concave since $h''(x) = 0$.

End Detour

EM Algorithm as max likelihood

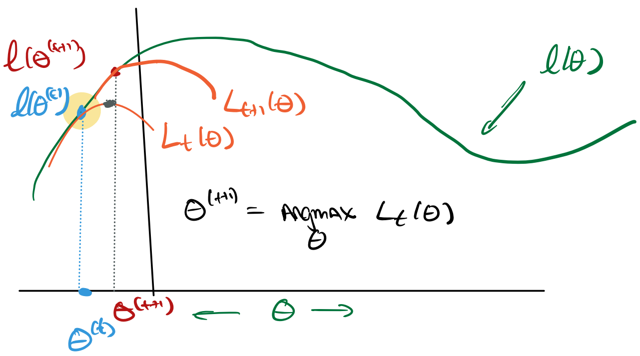
$$\ell(\theta) = \sum_{i=1}^{n} \log P(x^{(i)}; \theta)$$

PARAMETERS

DATA

We assume $P(x;\theta) = \sum_z P(x, z; \theta)$ of GMM latent variable

Picture of Algorithm



$L_t(\theta) \leq \ell(\theta)$ (lowerbound)

$L_t(\theta^{(t)}) = \ell(\theta^{(t)})$ (Tight)

Hope L_t easier to optimize than $\ell(\theta)$

Rough Algo

(E-step) 1. Given $\theta^{(t)}$ find L_t

(M-step) 2. Given L_t, set $\theta^{(t+1)} = \underset{\theta}{\text{argmax}}\ L_t(\theta)$

How do we construct L_t?

(let's look at single point)

$$\log \sum_z P(x, z; \theta) = \log \sum_z \frac{Q(z) P(x, z; \theta)}{Q(z)} \quad \text{for any } Q(z)$$

We pick $Q(z)$ s.t. $\sum_z Q(z) = 1$ AND $Q(z) \geq 0$ (*)

$$= \log \underset{z}{\mathbb{E}} \left[\frac{P(x, z; \theta)}{Q(z)} \right] \quad \text{(Symbol Pushing)}$$

$$\geq \underset{z}{\mathbb{E}}\left[\log \frac{P(x,z;\theta)}{Q(z)} \right]$$ JENSEN! (log IS CONCAVE)

$$= \sum_{z} Q(z) \log \frac{P(x,z;\theta)}{Q(z)}$$ (DEF of $\mathbb{E}$)

Key STEP holds for ANY SUCH Q! (*)

This gives a family of lower bounds, one for each choice of Q ($\ell_t \leq \ell$)

How do we make it tight? Select Q to make inequality tight

what if... $\log \frac{P(x,z;\theta)}{Q(z)} = c$ for some constant, then JENSEN'S IS Equality!

$P(x,z;\theta) = P(z|x;\theta) P(x;\theta)$

So, $Q(z) = P(z|x;\theta)$ then

$c = \log P(x;\theta)$ does not depend on z, so constant!

NB: $Q(z)$ does depend on θ & x – we will select a $Q^{(i)}(z)$ for every point independently.

WE DEFINE Evidence-BASED lower bound (ELBO), sum over z

$$\text{ELBO}(x, Q, z) = \sum_{z} Q(z) \log \frac{P(x,z;\theta)}{Q(z)}$$

WE'VE SHOWN $\ell(\theta) \geq \sum_{i=1}^{n} \text{ELBO}(x^{(i)}, Q^{(i)}; \theta)$ for ANY $Q^{(i)}$ satisfying (*)

lower bound

$$\ell(\theta^{(t)}) = \sum_{i=1}^{n} \text{ELBO}(x^{(i)}, Q^{(i)}, \theta^{(t)})$$ for choice of $Q^{(i)}$ above.

WRAP UP

1. (E-STEP) $Q^{(i)}(z) = P(z^{(i)} \mid x^{(i)}; \theta)$ for $i=1 \dots n$
2. (M-STEP) $\theta^{(t+1)} = \underset{\theta}{\text{ARGMAX}}\ L_t(\theta)$

in which $L_t(\theta) = \sum_{i=1}^{n} \text{ELBO}(x^{(i)}, Q^{(i)}; \theta)$

WHY DOES this TERMINATE? $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$

IS IT Globally OPTIMAL? (NOPE SEE PICTURE)

WE DERIVED HARD & SOFT CLUSTERING METHOD

EM Algorithm IN TERMS of MLE.

EM for Mixture of Gaussians

Restate EM

(E-STEP) for $i = 1..n$ SET $Q_i(z) = P(z^{(i)} \mid x^{(i)}, \theta^{(t)})$

↳ DATA & INPUT PARAMS

(M-STEP) $\theta^{(t+1)} = \operatorname{Argmax}_{\theta} \mathcal{L}_t(\theta)$

$= \operatorname{Argmax}_{\theta} \sum_{i=1}^{n} ELBO(x^{(i)}, Q^{(i)}, \theta)$

Warm Up: Mixture of Gaussians.

EM RECOVERS OUR AD HOC Algorithm

$$P(x^{(i)}, z^{(i)}) = P(x^{(i)} \mid z^{(i)}) P(z^{(i)})$$

$z^{(i)} \sim \text{Multinomial}(\Phi)$ $\quad \phi_i \geq 0 \quad \sum \phi_i = 1$ "IN cluster j"

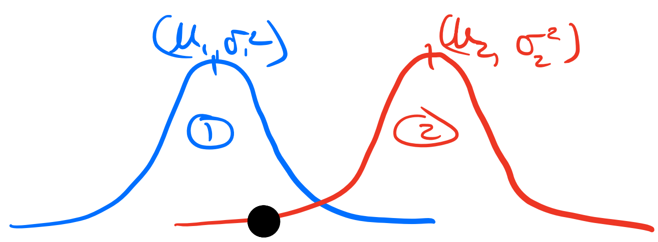
$x^{(i)} \mid z^{(i)} = j \sim N(\mu_j, \sigma_j^2)$ "CLUSTER MEANS"

$z^{(i)}$ IS OUR LATENT VARIABLE.

WHAT IS EM HERE?

$$Q_i(z) = P(z^{(i)} = j \mid x^{(i)}; \theta)$$

WE SAW THAT could compute VIA Bayes Rule $P(x^{(i)} \mid z^{(i)} = j; \theta)$



1. much more likely for ① than ②
2. But if we knew, $\phi_2 >> \phi_1$ maybe we'd think likely from ②

Bayes Rule AUTOMATES this REASONING

M-STEP: Compute DERIVATIVES...

$$\max_{\phi, \mu, \Sigma} \sum_{i=1}^{n} \underbrace{\sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}}_{f_i(\theta)}$$

WRITE θ FOR NOTATION ABOVE

RECALL $\quad w_j^{(i)} \triangleq Q_i(z=j)$

$$P(x^{(i)}, z^{(i)}; \theta) = P(x^{(i)} \mid z^{(i)}) P(z^{(i)})$$

GAUSSIAN (μ, Σ)

$$f_i(\theta) = \sum_j w_j^{(i)} \log\left(\frac{\frac{1}{2\pi |\Sigma_j|^{1/2}} \exp\left\{-\frac{1}{2}(x^{(i)}-\mu_j)^T \Sigma_j^{-1} (x^{(i)}-\mu_j)\right\} \cdot \phi_j}{w_j^{(i)}} \right)$$

$$\nabla_{\mu_j} \sum_i f_i(\theta) = \sum_i \nabla_{\mu_j} \left(w_j^{(i)} \; -\frac{1}{2}(x^{(i)}-\mu_j)^T \Sigma_j^{-1} (x^{(i)}-\mu_j) \right)$$

$$= -\frac{1}{2} \sum_i w_j^{(i)} \Sigma_j^{-1} (x^{(i)}-\mu_j) = -\frac{1}{2} \Sigma_j^{-1} \left(\sum_i w_j^{(i)} (x^{(i)}-\mu_j) \right)$$

SETTING TO 0 AND USING Σ_j^{-1} IS FULL RANK $\Rightarrow \sum_i w_j^{(i)} (x^{(i)}-\mu_j) = 0$

$$\therefore \mu_j = \frac{\sum_i w_j^{(i)} x^{(i)}}{\sum_i w_j^{(i)}} \quad \text{(AS BEFORE)}$$

ϕ_j IS CONSTRAINED $\sum_j \phi_j = 1$, $\phi_j \geq 0$, NEED LAGRANGIAN

$$\nabla \phi_j = \sum_{i=1}^{n} w_j^{(i)} \nabla_{\phi_j} \log \phi_j + \nabla_{\phi_j} \lambda \left(\sum_j \phi_j - 1 \right)$$

$$= \sum_{i=1}^{n} \frac{w_j^{(i)}}{\phi_j} + \lambda = 0 \Rightarrow \phi_j = -\frac{1}{\lambda} \sum_{i=1}^{n} w_j^{(i)}$$

SINCE $\sum \phi_j = 1$, $\quad \sum_j \phi_j = -\frac{1}{\lambda} \sum_{i,j} w_j^{(i)} = -\frac{n}{\lambda}$

$$\therefore \phi_j = \frac{1}{n} \sum_i w_j^{(i)}$$

MESSAGE: EM RECOVERS GMM AUTOMATICALLY.

NB: IF $z^{(i)}$ IS CONTINUOUS, CAN CAN REPLACE SUMS W/ INTEGRALS

Factor Analysis

Many fewer points than dimensions "$n \ll d$"

cf: GMMs $n \gg d$ lots of photons, few sources.

How does this happen?

Place sensors all over campus, record @ 1000s of locations

$\Rightarrow d \sim 1000s$

But only record for 30 days $(n < d)$

Want to fit a density but seems hopeless.

KEY IDEA: Assume there is some latent r.v. that is not too complex and explains behavior.

1st let's see problems w/ GMMs.... Even 1 Gaussian

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x^{(i)} \longrightarrow \text{this is OK}$$

$$\Sigma = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \mu)(x^{(i)} - \mu)^T$$

$\text{Rank}(\Sigma) \leq n < d$ – not full rank.

Problem in Gaussian likelihood

$$P(x; \mu, \Sigma) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp \left\{ (x-\mu)^T \Sigma^{-1} (x-\mu) \right\}$$

Σ^{-1} is not defined.

$|\Sigma| = 0$

We will fix these issues by examining three models that are simpler. Spoiler: we'll combine these in the end!

Recall MLE for Gaussian

$$\max_{\mu,\Sigma_i} \sum_{i=1}^{n} \log \frac{1}{2\pi |\Sigma_i|^{1/2}} \exp\left\{-\tfrac{1}{2}(x-\mu)^T \Sigma_i^{-1}(x-\mu)\right\}$$

Equivalent

$$\min_{\mu,\Sigma_i} \sum_{i=1}^{n} (x-\mu)^T \Sigma_i^{-1}(x-\mu) + \log|\Sigma_i|$$

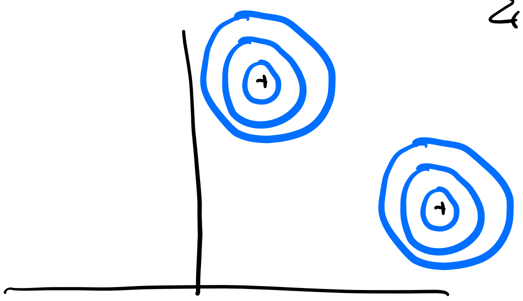
If Σ_i is full rank, $\nabla_\mu = \sum_{i=1}^{n} \Sigma_i^{-1}(x-\mu) = 0 \Rightarrow \mu = \frac{1}{n}\sum_i x^{(i)}$

we'll use this as plugin below.

Building Block 1

Suppose independent and identical covariance

$\Sigma_i = \sigma^2 I$ (NB: Parameter Tying)



Covariance "are circles"

What is MLE for Σ_i? $|\Sigma_i| = z^d$

$$\min_{\sigma^2} \sigma^{-2} \underbrace{\sum_{i=1}^{n}(x^{(i)}-\mu)^T(x-\mu)}_{c} + d\log\sigma^2$$

let $z=\sigma^2$ $\quad \min_z \frac{1}{z} c + d\log z$

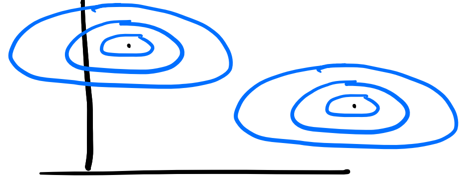
$$\Rightarrow \nabla_z = -z^{-2}c + \frac{nd}{z} = 0 \Rightarrow z = \frac{c}{nd}$$

$$\therefore \sigma^2 = \frac{1}{nd}\sum_{i=1}^{n}(x^{(i)}-\mu)^T(x^{(i)}-\mu)$$

"Substract mean and square all entries."

Building Block 2

$$\Sigma = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_d^2 \end{bmatrix}$$



Axis Aligned ellipse

SET $z_i = \sigma_i^2$ (SAME IDEA AS ABOVE)

$$\min_{z_1 \ldots z_d} \sum_{i=1}^{n} \sum_{j=1}^{d} z_j^{-1}(x^{(i)} - \mu_j)^2 + \log z_j$$

this is d Problems for Each 1 dimension

$$\Rightarrow \sum_{i=1}^{n} z_j^{-1}(x^{(i)} - \mu_j)^2 + \log z_j$$

$$\Rightarrow \sigma_j^2 = \frac{1}{n}\sum_i (x_j^{(i)} - \mu_j)^2$$

Our FACTOR model

PARAMETERS

$\mu \in \mathbb{R}^d$

$\Lambda \in \mathbb{R}^{d \times s}$

$\Phi \in \mathbb{R}^{d \times d}$ — DIAGONAL MATRIX

MODEL

$$P(x, z) = P(x | z) P(z)$$ z IS LATENT

$z \sim N(0, I) \in \mathbb{R}^S$ for $s < d$ "small dim"

$x = \mu + \Lambda z + \epsilon$ OR $x \sim N(\mu + \Lambda z, \Phi)$

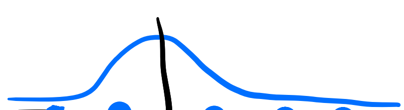
MEAN IN the SPACE

$\hookrightarrow$ MAPS FROM SMALL LATENT SPACE TO LARGE SPACE

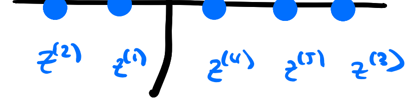
$\epsilon \sim N(0, \Phi)$ Noisy

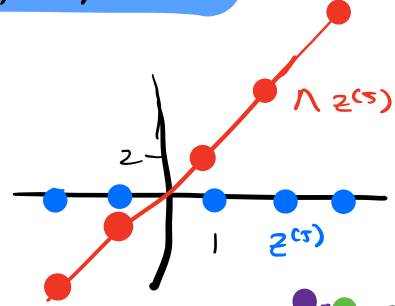
Ex: $d = 2$, $s = 1$, $n = 5$ $x = \mu + \Lambda z + \epsilon$

1. GENERATE $z^{(1)}, \ldots, z^{(5)}$ from $N(0, 1)$



2. Suppose $\Lambda = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

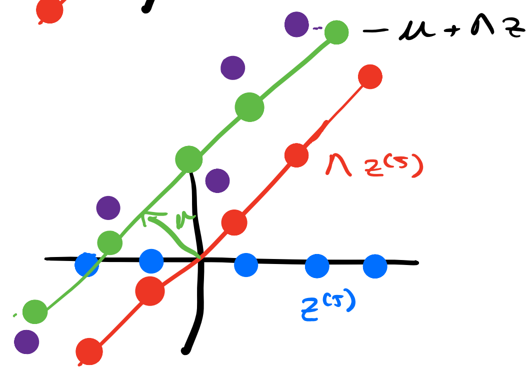




3. Add μ

4. Add ϵ

↳ Big space



Data we would observe are purple dots

So small latent space produces data in high dim space.

<u>Technical tools</u>: Block Gaussians

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad x_1 \in \mathbb{R}^{d_1}, \; x_2 \in \mathbb{R}^{d_2}$$

$$x \in \mathbb{R}^{d}$$

$$\Sigma = \begin{array}{cc} \begin{array}{cc} d_1 & d_2 \end{array} & \\ \begin{bmatrix} \Sigma_{11} & \Sigma_{21} \\ \Sigma_{12} & \Sigma_{22} \end{bmatrix} & \begin{array}{l} \}d_1 \\ \}d_2 \end{array} \end{array} \quad \Sigma_{ij} \in \mathbb{R}^{d_i \times d_j} \qquad i,j \in \{1,2\}$$

Notation is widely used and helpful.

<u>Fact 1:</u> $P(x_1) = \int_{x_2} P(x_1, x_2)$ Marginalization

For Gaussians, $p(x_1) = N(\mu_1, \Sigma_{11})$ (not surprising)

<u>Fact 2:</u> $P(x_1 \mid x_2) \sim N(\mu_{1|2}, \Sigma_{1|2})$ Conditioning

$$\mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \qquad \text{(MATRIX INVERSION LEMMA)}$$

Proofs outline (HAPPY TO ADD)

Summary: MARGINALIZATION & CONDITIONING GAUSSIAN $\Rightarrow$ ANOTHER GAUSSIAN (CLOSED)

WE HAVE FORMULA FOR PARAMETERS.

BACK TO FACTOR ANALYSIS

$$x = \mu + \Lambda z + \epsilon$$

$$\begin{pmatrix} z \\ x \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ \mu \end{pmatrix}, \Sigma \right)$$

SINCE $\mathbb{E}[z] = 0$

$\mathbb{E}[x] = \mu$

WHAT IS Σ?

$$\Sigma_{11} = \mathbb{E}[zz^T] = I$$

$$\Sigma_{12} = \mathbb{E}[z(x-\mu)^T] = \mathbb{E}[zz^T\Lambda^T] + \cancel{\mathbb{E}[z\epsilon^T]}^{\,0}$$

$$= \Lambda^T$$

$$\Sigma_{21} = \Sigma_{12}^T$$

$$\begin{aligned}
\Sigma_{22} &= \mathbb{E}[(x-\mu)(x-\mu)^T] \\
&= \mathbb{E}[(\Lambda z + \epsilon)(\Lambda z + \epsilon)^T] \\
&= \mathbb{E}[\Lambda z z^T \Lambda^T] + \mathbb{E}[\epsilon\epsilon^T] \\
&= \Lambda\Lambda^T + \Phi
\end{aligned}$$

$$\Sigma = \begin{bmatrix} I & \Lambda^T \\ \Lambda & \Lambda\Lambda^T + \Phi \end{bmatrix}$$

E-STEP: $Q_i(z) = P(z^{(i)} \mid x^{(i)}; \theta)$ — USE CONDITIONAL!

M-STEP: WE HAVE CLOSED FORMS!

Summary:

- WE SAW that EM CAPTURES GMM
- WE LEARNED ABOUT FACTOR Analysis (LATENT low dim. STRUCTURE)
- WE SAW HOW to ESTIMATE PARAMETERS OF FA using EM.

PCA & ICA

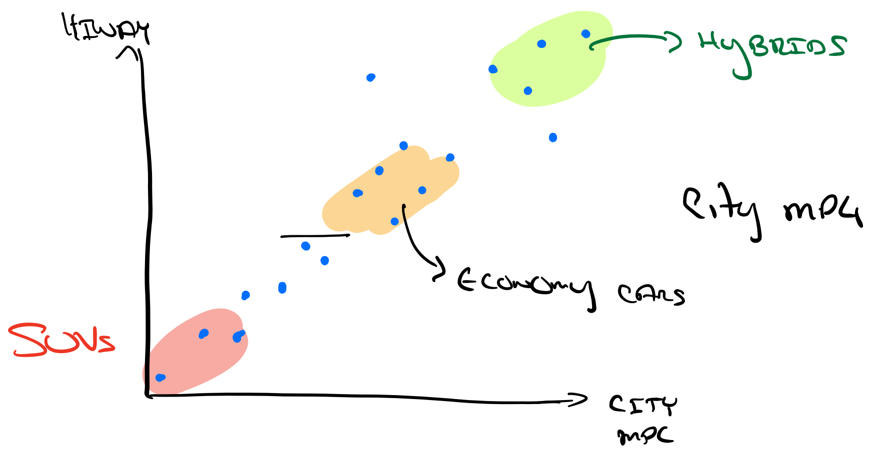
+ Recap PCA & solve it
+ ICA & "the cocktail party" "learn upto symmetry"

UP Next: Self-Supervised Machine Learning!

PCA: Principal Component Analysis

STRUCTURE	Prob.	Non Prob
"Cluster"	GMM	K-MEANS
"Subspace"	Factor Analysis	PCA ← THIS SECTION

Ex: Given pairs (Hiway mpg, City mpg) of some cars

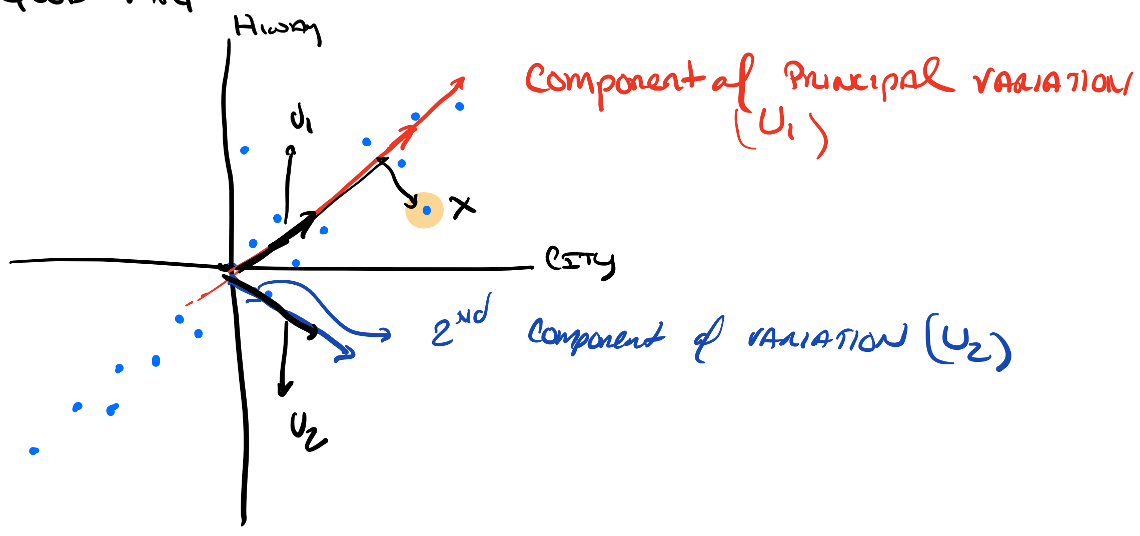


Question: "Good mpg"

① CENTER DATA

$\mu = \frac{1}{n} \sum_i x^{(i)}$

$x^{(i)} \mapsto x^{(i)} - \mu$



Now $\|u_1\| = \|u_2\| = 1$ by convention.

- u_1 is "How good is mpg"
- u_2 is "difference between Hiway & city" (Roughly)

We can write $x = \alpha_1 u_1 + \alpha_2 u_2$

↳ We may just keep this component

"Explains more variation"

TODAY: How we find these directions, and some caveats

- think about 1000s of dims $\longrightarrow$ 10s of dims
- A dimensionality reduction method

Preprocessing

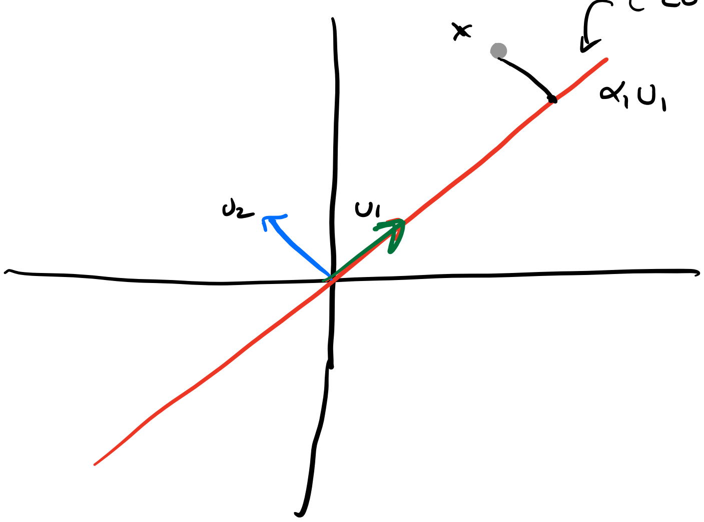
Given $x^{(1)} \dots x^{(n)} \in \mathbb{R}^d$

1. Center the data $x^{(i)} \mapsto x^{(i)} - \mu$ in which $\mu = \frac{1}{n}\sum_i x^{(i)}$
2. May need to rescale components e.g. "feet per gallon" & MPG

We will assume data is preprocessed

PCA as Optimization

$\{t u_1 : t \in \mathbb{R}\}$



$\alpha_1 u_1$ closest poin to line

We have (1) $\|u_i\| = 1$ (unit vectors)

(2) $u_i \cdot u_j = \delta_{ij}$ (orthogonal)

How do you find closest point to the line?

$$\alpha_1 = \arg\min_{\alpha} \|x - \alpha u_1\|^2$$

$$= \arg\min_{\alpha} \|x\|^2 + \alpha^2 \cancel{\|u_i\|^2}^{\,1} - 2\alpha(u_i \cdot x)$$

differentiate wrt α $\quad 2(\alpha - u_i \cdot x) = 0 \Rightarrow \alpha = u_i \cdot x$

PCA & ICA

+ Recap PCA & solve it

+ ICA & "the Cocktail Party" "Learn upto Symmetry"

UP Next: Self-Supervised Machine Learning!

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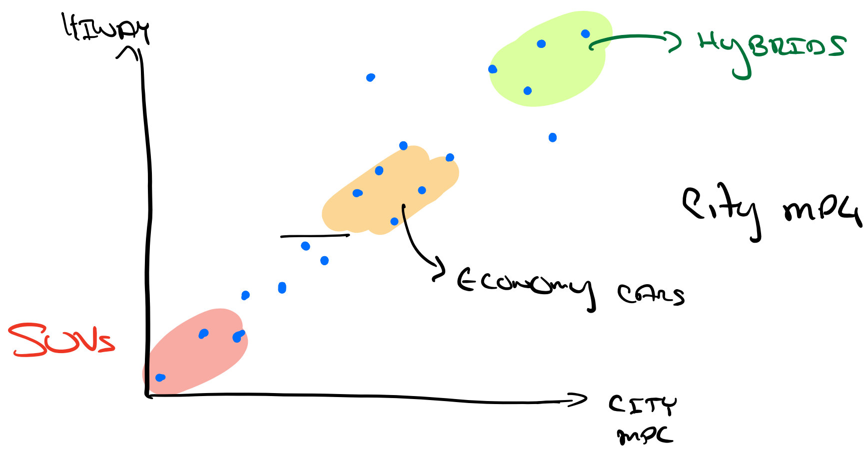
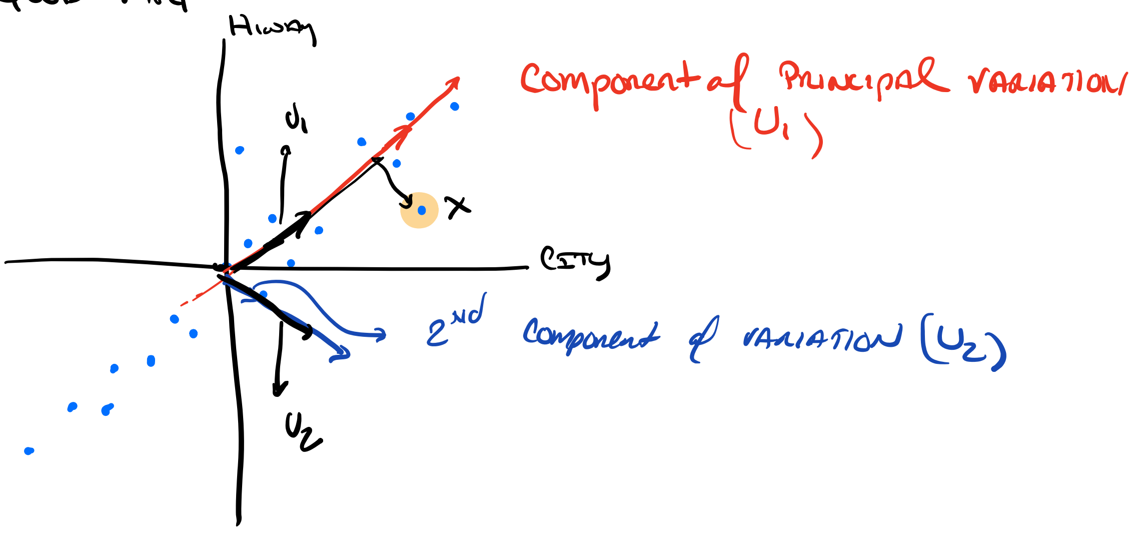


Question: "Good mpg"

① Center data

$$\mu = \frac{1}{n} \sum_i x^{(i)}$$

$$x^{(i)} \mapsto x^{(i)} - \mu$$



Now $\|u_1\| = \|u_2\| = 1$ by convention.

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We can write $x = \alpha_1 U_1 + \alpha_2 U_2$

↳ We may just keep this component

"EXPLAINS MORE VARIATION"

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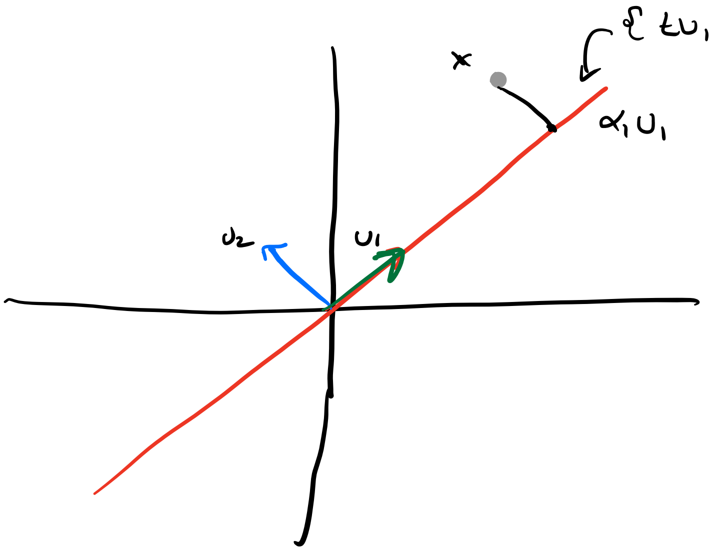
We will assume data is preprocessed

PCA as optimization



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$$= \operatorname*{argmin}_{\alpha} \|x\|^2 + \alpha^2 \cancel{\|u_1\|^2}^{\,1} - 2\alpha (u_1 \cdot x)$$

differentiate wrt α $\quad 2(\alpha - u_1 \cdot x) = 0 \Rightarrow \alpha = u_1 \cdot x$

Generalize: $u_1 \cdots u_k \in \mathbb{R}^d$ AND $x \in \mathbb{R}^d$

USE $u_i \cdot u_j = \delta_{ij}$

$$\underset{\alpha_1 \cdots \alpha_d}{\operatorname{Argmin}} \|x - \sum_{i=1}^{k} \alpha_i u_i\|^2 = \underset{\alpha}{\operatorname{argmin}} \|x\|^2 + \sum_{i=1}^{n} \alpha_i^2 \|u_i\|^2 - 2\alpha_i \langle u_i \cdot x \rangle$$

Hence $\alpha_i = u_i \cdot x$

WE CALL $\|x - \sum_{i=1}^{k} \alpha_i u_i\|^2$ the RESIDUAL

WE CAN find PCA by either

IN class ① MAXIMIZE Projected Subspace

② MINIMIZE Residual

$$\max_{\substack{u \in \mathbb{R}^d \\ \|u\| = 1}} \frac{1}{n} \sum_{i=1}^{n} (u \cdot x^{(i)})^2$$

WE NEED some facts to solve this

LET A be symmetric & square, then

$A = U \Lambda U^T$ IN which

- $UU^T = I$ (ORTHONORMAL)
- Λ is diagonal

$\Lambda_{ii} = \lambda_i$ AND $\lambda_1 \geq \cdots \geq \lambda_n$ by convention

eigenvalues

Recall: If $x = \sum_{i=1}^{n} \alpha_i u_i$ where $[u_1 \cdots u_n] = U$

STANDARD BASIS vector

$$Ax = U\Lambda U^T x = U\Lambda \sum_{i=1}^{n} \alpha_i e_i \qquad (u_i \cdot u_j = \delta_{ij})$$

$$= U \sum_{i=1}^{n} \lambda_i \alpha_i e_i \qquad \text{diagonal } \Lambda$$

$$= \sum \lambda_i \alpha_i u_i$$

If $x = c u_i$ then x is AN eigenvector, AND $Ax = \lambda_i x$

$$\max_{x: \|x\|^2 = 1} x^T A x = \max_{\alpha: \|\alpha\|^2=1} \sum_{i=1}^{n} \alpha_i^2 \lambda_i$$

Hence, we set $\alpha_i = 1$, the principal eigenvalue

Which x attains it? If $\lambda_1 = \lambda_2$?

Now, back to PCA!

THE projection onto u

$$\max_{\substack{u: \|u\|^2=1 \\ u \in \mathbb{R}^d}} \frac{1}{n} \sum_{i=1}^{n} (u_i \cdot x^{(i)})^2$$

$$= \frac{1}{n} \sum_{i=1}^{n} u^T x^{(i)} (x^{(i)})^T u = u^T \Big(\frac{1}{n} \sum x^{(i)} (x^{(i)})^T \Big) u$$

Covariance of data (we subtracted mean)

$\therefore$ u is principal eigenvector

What if we want more dimensions? we keep top-k!

How do we represent data?

$$x^{(i)} \mapsto \sum_{j=1}^{k} (x^{(i)} \cdot u_j) u_j$$

we keep these k scalars (the α_i above)

A map from $\mathbb{R}^d \to \mathbb{R}^k$

How do we choose k?

One approach "Amount of Explained Variance"

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{n} \lambda_i} \geq 0.9$$

(aside $tr(A) = \sum_i A_{ii} = \sum_j \lambda_j$)

$j=1$

NB: Only makes sense if $\lambda_j \geq 0$. Hence covariance is important

Lurking instability: Suppose $\lambda_k = \lambda_{k+1} \ldots$ what happens?

Rep is unstable here

Recap of PCA

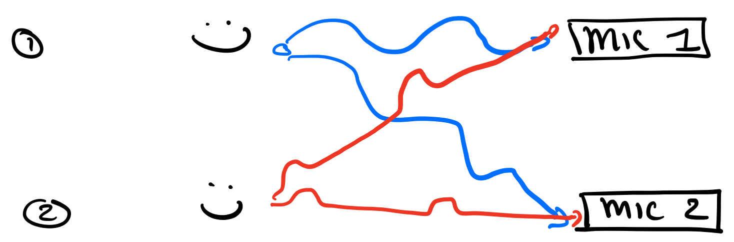
- Dimensionality reduction technique (e.g. Visualization)
- Main idea is to project on a subspace, nice theory.

ICA Independent Component Analysis

- high-level story
- Key facts & likelihood
- model

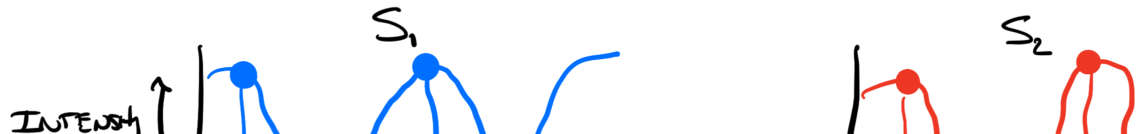
Cocktail Party Problem (in HW!)

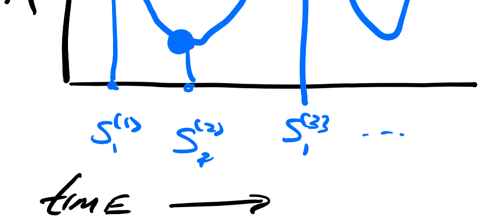
People Microphones

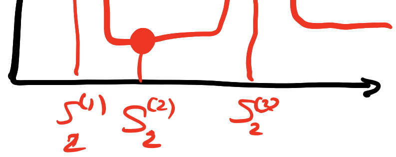


NB: We see a mixture of ~ & ~ at each mic

Speaker S_1, S_2 Data X







$s_j^{(t)}$ IS <u>INTENSITY</u> AT <u>TIME</u> t from <u>SPEAKER</u> j

WE DO <u>NOT</u> OBSERVE $s^{(t)}$ only $x^{(t)}$ – the microphones

<u>ex</u> model $\quad x_j^{(t)} = a_{j1} s_1^{(t)} + a_{j2} s_2^{(t)}$

"microphone j sees a mixture of $s_1^{(t)}$ & $s_2^{(t)}$"

LATENT

<u>OR</u> $\quad x^{(t)} = A s^{(t)}$

OBSERVED $\quad$ LATENT

for simplicity, assume # of speakers = # of mics = d

<u>GIVEN</u>: $x^{(1)}, x^{(2)}, \ldots, x^{(n)} \in \mathbb{R}^d$ $\qquad$ d is # of microphones & speakers

<u>DO</u>: find $s^{(1)}, \ldots, s^{(n)} \in \mathbb{R}^d$

AND $A \in \mathbb{R}^{d \times d}$ s.t. $x^{(t)} = A s^{(t)}$

WE CALL A the mixing matrix AND $W = A^{-1}$ unmixing matrix

WRITE $W = \begin{bmatrix} w_1^T \\ \vdots \\ w_d^T \end{bmatrix}$ so that $s_j^{(t)} = w_j \cdot x^{(t)}$

<u>Some Caveats</u>

- WE ASSUME A does not vary w/ time AND is full RANK

- THERE ARE INHERENT Ambiguity
  - WE CAN'T DETERMINE SPEAKER ID (could swap 1 & 2)
  - CAN'T DETERMINE Absolute INTENSITY

$$(cA)(c^{-1}s^{(t)}) = As^{(t)} \quad \text{for any } c \neq 0$$

- Surprising Speakers cannot be Gaussian

Suppose so $\quad x^{(i)} \sim N(\mu, AA^T)$ then if $U^TU = I$ $\quad AU$ generates the SAME data.

Nevertheless, we can recover something meaningful!

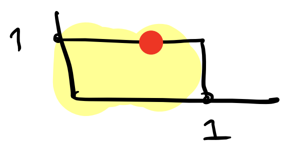
Algorithm: Just MLE, solved by Grad Descent

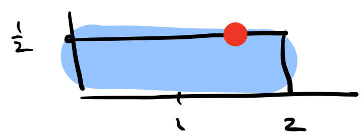
Detour: Density under linear transform (Key Confusion)

Ex: $S \sim \text{Uniform}[0,1]$ $\quad U = 2S$ $\quad$ what is PDF of U?

Tempted to write $P_U\left(\frac{x}{2}\right) = P_S(x)$

PDF of S





PDF of U

$$P_S(x) = \begin{cases} 1 & \text{if } x \in [0,1] \\ 0 & \text{o.w} \end{cases} \qquad P_U(x) = P_S\left(\frac{x}{2}\right) \cdot \frac{1}{2}$$

The key issue is the Normalization constant

for invertible matrix A, $\quad U = AS$

$$P_U(x) = P_S(A^{-1}x)\,|\det(A^{-1})|$$

$$= P_S(Wx)\,|\det(W)| \qquad \left(\frac{1}{\det(A)} = \det(A^{-1})\right)$$

Change of var formula for integrals

From here ICA is MLE:

$$P(s) = \prod_{j=1}^{d} P_s(s_j)$$ "sources are independent, and have same distribution"

$$P(x) = \prod_{j=1}^{d} P_s(w_j \cdot x) \cdot |\det(W)|$$ (use linear transform above)

Now written in terms of x and A.

Key technical bit: Use non-rotationally invariant distribution

SET $P_s(x) \propto g'(x)$ for $g(x) = (1+e^{-x})^{-1}$

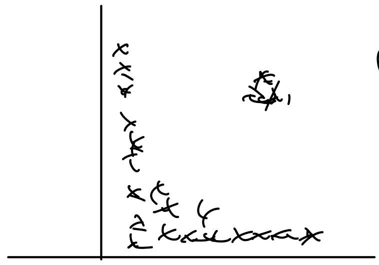
Solve $$\ell(W) = \sum_{i=1}^{n} \sum_{j=1}^{d} \log g'(w_j \cdot x^{(i)}) + \log|\det(W)|$$

- $\log|\det(W)|$
- Use GD & you're done!

Recap: · Saw PCA. Workhorse dimensionality reduction

· ICA. Key ideas for HW. Introduce "up to symmetry".

The EM Algorithm



Recall Unsupervised Algorithms
K-means & GMM

Big Idea latent variable → "fraction of points from a source"

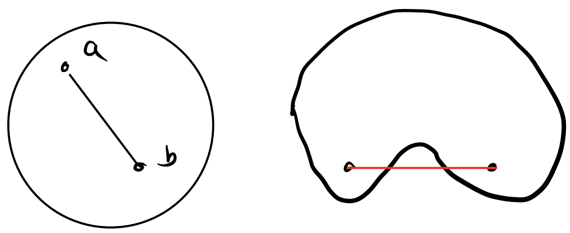
Algorithm "guess" latent variable
Estimate the parameters (center and shape)

Today: EM Algorithm for latent variable models

+ Technical Detour: Convexity & Jensen's Inequality
+ EM Algorithm as MLE
+ GMM as EM Algorithm
+ Factor Analysis (EM for a new setting)

Detour Convexity & Jensen (This is a key result, we'll go slowly)

A set Ω is convex if for any $a,b \in \Omega$ the line joining a,b is in Ω as well.



Convex

NOT convex!

(need to check $\forall a,b \in \Omega$)

In symbols,

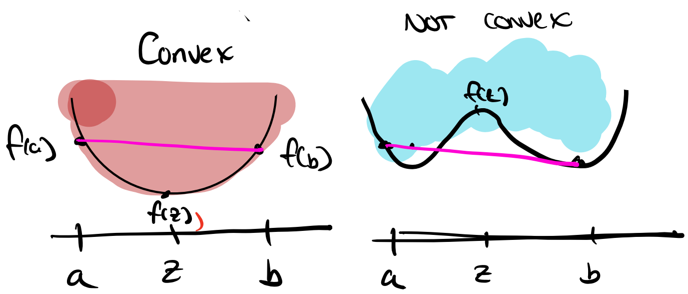
$$\forall \lambda \in [0,1], \quad a,b \in \Omega$$

$$\lambda a + (1-\lambda) b \in \Omega$$

Given a function f, the graph of f G_f is defined as

$$G_f = \{ (x,y) : y \geq f(x) \}$$

A function is convex if its graph is convex (as a set)



"Every chord is above function"

In symbols, $\forall \lambda \in [0,1]$

$$\lambda (a, f(a)) + (1-\lambda)(b, f(b)) \in \Omega$$

or let $z = \lambda a + (1-\lambda) b$

$$\lambda f(a) + (1-\lambda) f(b) \geq f(z)$$

If f twice differentiable, $\forall x$ $f''(x) \geq 0 \Rightarrow f$ is convex

Pf
$$f(a) = f(z) + f'(z)(a-z) + f''(\xi_a)(a-z)^2 \qquad \xi_a \in [a,z]$$

$$f(b) = f(z) + f'(z)(b-z) + f''(\xi_b)(b-z)^2 \qquad \xi_b \in [z,b]$$

$$\lambda f(a) + (1-\lambda) f(b) = f(z) + f'(z)(\overset{0}{\cancel{\lambda a + (1-\lambda) b - z}}) + c \qquad c \geq 0$$

def of z.

i.e. $\lambda f(a) + (1-\lambda) f(b) \geq f(z)$ □

We say f is strongly convex if $\forall x \in \mathrm{Dom}(f)$ $f''(x) > 0$.

Ex: $f(x) = x^2 \Rightarrow f''(x) = 2 \Rightarrow$ strongly convex

$f(x) = x^2(x-1)^2$: graph above (not convex)

Jensen's Inequality $\mathbb{E}[f(x)] \geq f(\mathbb{E}[x])$ for convex f.

Ex: x takes value a with prob λ

takes value b with prob $1-\lambda$

$\mathbb{E}[f(x)] = \lambda f(a) + (1-\lambda) f(b)$

$f(\mathbb{E}[x]) = f(z)$ $\quad z = \lambda a + (1-\lambda) b$

NB: can prove finitely supported distribution by induction

for convex f, definition implies thm in this case!

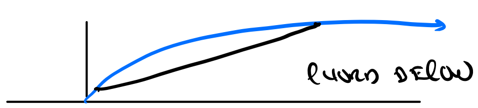
Stronger if f is strongly convex, and $\mathbb{E}[f(x)] = f(\mathbb{E}[x])$

$\Rightarrow X$ is a constant (experts: almost surely)

We need concave functions g concave iff $-g$ is convex

$\mathbb{E}[g(x)] \leq g(\mathbb{E}[x])$

ex: $g(x) = \log(x) \Rightarrow g''(x) = -x^{-2}$ on $(0, \infty)$ negative



What about $h(x) = ax + b$ convex & concave since $h''(x) = 0$.

End Detour